PROJECT DOCUMENTATION | DATA ANALYTICS

Factory Reallocation & Shipping Optimization Recommendation System

for Nassau Candy Distributor

SQL Python Power BI DAX GitHub

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Project Overview & Objective

PROJECT OBJECTIVE

To analyze sales, profitability, shipping performance, lead time and shipping risk in order to identify opportunities for reducing shipping costs and improving operational efficiency for Nassau Candy Distributor.

FOUR KEY BUSINESS OBJECTIVES

1

Profitability Analysis

Find which products, divisions and regions drive or reduce profit.

2

Shipping Performance

Find which shipping methods and regions increase lead time.

3

Shipping Risk Analysis

Identify the regions where shipping risk is highest.

4

Optimization Recommendations

Recommend actions to improve shipping efficiency.

PROBLEM STATEMENT

Business Problems & Key Questions

Problem 1: Profitability Analysis

PROBLEM

Which products and regions increase profit, and where can profitability be improved?

KEY QUESTIONS

Total sales, gross profit and margin? Top product, division and region? Which products have low margins?

Problem 2: Shipping Performance

PROBLEM

How do shipping methods and regions change delivery lead time?

KEY QUESTIONS

Average lead time? Slowest and fastest ship mode? Slow-category orders? Is risk linked to lead time?

Problem 3: Shipping Risk Analysis

PROBLEM

Which places carry the highest shipping risk?

KEY QUESTIONS

How many and what % of orders are High risk? Which regions, modes and products are most exposed?

Problem 4: Factory Reallocation (Core)

PROBLEM

Can distribution or factory allocation be changed to cut shipping cost and lead time?

KEY QUESTIONS

Which regions cost most? High lead time + high risk? Which regions to prioritize? What extra data is needed?

Dataset Description

CATEGORY	COLUMNS	PURPOSE
Order Details	Order ID, Order Date, Ship Date	Order analysis
Product Details	Product ID, Product Name, Division	Product performance
Location	City, State/Province, Region	Regional analysis
Financial	Sales, Cost, Gross Profit, Profit Margin	Profitability
Shipping	Ship Mode, Lead Time, Shipping Risk	Shipping performance
Quantity	Units	Product volume
Time	Order Year, Order Month, Order Quarter	Monthly & quarterly trends
Additional	Lead Time Category	Delivery performance classification

METHODOLOGY

Project Workflow

STAGE 1

SQL

1

Import, data types, data quality checks

STAGE 2

SQL

2

Cleaning and business queries

STAGE 3

Python

3

EDA and optimization analysis

STAGE 4

Power BI

4

Four-page interactive dashboard

STAGE 5

Power BI

5

DAX measures and insights

STAGE 6

GitHub

6

SQL scripts, Python notebook, README, dashboard screenshots

SQL: Data Validation & Business Queries

DATA QUALITY CHECKS

- Total record count to confirm a complete import
- Data type verification for dates, numeric and text columns
- Missing values and duplicate Order IDs
- Date logic: Ship Date must not be earlier than Order Date
- Financial consistency: Gross Profit = Sales - Cost, and margin = Gross Profit / Sales

CLEANING & BUSINESS QUERIES

- Sales and profit by product, division, region
- Lead time by ship mode and region
- High-risk order counts and percentage

SAMPLE CHECK 1: TOTAL RECORDS

```
SELECT COUNT(*) AS Total_Records
FROM your_table;
```

Replace your_table with the imported table name.

STAGES 4 AND 5

Power BI Dashboard: 4 Pages

Page 1 Executive Overview

- KPIs: Sales, Gross Profit, Margin %, Orders, Avg Lead Time
- Monthly sales trend, profit by division
- Sales by region, shipping risk distribution
- Slicers: Year, Quarter, Region, Division, Ship Mode

FOCUS: Business snapshot at a glance

Page 2 Sales & Profitability

- Sales and gross profit by product
- Profit by region, sales by division
- Product profitability table

FOCUS: Which products and regions generate the highest profit?

Page 3 Shipping & Risk

- Lead time by ship mode and by region
- Shipping risk by region and by ship mode
- Lead time category distribution

FOCUS: Which regions and shipping methods need improvement?

Page 4 Optimization & Recommendations

- Cost by region, lead time vs profit scatter
- Regional performance matrix, high-risk ranking
- Business recommendation table

FOCUS: Where to prioritize shipping-efficiency improvements?

Key DAX Measures

MEASURE	DAX FORMULA	PURPOSE
Total Sales	`SUM('Nassau Candy Distributor'[Sales])`	Total sales revenue
Total Gross Profit	`SUM('Nassau Candy Distributor'[Gross Profit])`	Profit after recorded costs
Profit Margin %	`DIVIDE([Total Gross Profit], [Total Sales])`	Overall profitability
Total Orders	`DISTINCTCOUNT('Nassau Candy Distributor'[Order ID])`	Order volume
Avg Lead Time	`AVERAGE('Nassau Candy Distributor'[Lead Time])`	Delivery speed
High-Risk Orders	`CALCULATE(COUNTROWS('Nassau Candy Distributor'), 'Nassau Candy Distributor'[Shipping Risk] = "High")`	Shipping exposure
High-Risk %	`DIVIDE([High-Risk Orders], COUNTROWS('Nassau Candy Distributor'))`	Share of risky orders

CORE OF THE PROJECT

Factory Reallocation: Two-Level Approach

LEVEL 1: Existing Dataset Only

Analysis possible with the current data:

- Regional cost analysis
- Lead time analysis
- Shipping risk analysis
- Product profitability
- Shipping-efficiency improvement recommendations

LEVEL 2: Factory Reallocation Optimization

Additional columns required for allocation recommendations:

Factory ID	Factory identification
Factory Location	Factory location
Factory Capacity	Production capacity
Available Capacity	Remaining capacity
Customer Demand	Customer demand
Shipping Distance	Factory-to-customer distance
Shipping Cost	Actual shipment cost
Production Cost	Factory-level production cost

CONCLUSION

Expected Outcomes & Deliverables

BUSINESS OUTCOMES

- Identify high-profit products and low-margin products to improve
- Pinpoint shipping methods and regions with long lead times
- Rank regions by shipping risk and cost
- Prioritize regions for shipping-efficiency improvements
- Define the factory capacity and location data needed to recommend reallocation

TAKEAWAY

Use the dataset to guide shipping-efficiency actions now, then add factory-level data to turn them into reallocation decisions.

GITHUB DELIVERABLES

- SQL scripts (checks, cleaning, business queries)
- Python notebook (EDA and optimization)
- Power BI dashboard (4 pages) and screenshots
- README with project documentation